Hailiang Zhao

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ZJU 100 Young Professor, School of Software Technology, Zhejiang University
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Research Profile

I work on algorithms and systems for intelligent services, with current interests in learning-augmented systems, agentic services, generative AI runtime, and cloud-edge intelligence. My research connects online optimization, machine learning, and systems design. Related methods have been evaluated in deployments involving domestic intelligent-computing chips and enterprise AI services.

Academic Appointments

Zhejiang University, School of Software Technology Aug. 2024 - Present
ZJU 100 Young Professor; Ph.D. Supervisor

Education and Visiting Research

Zhejiang University, College of Computer Science and Technology 2019 - 2024
Ph.D. in Computer Science and Technology; Advisor: Prof. Shuiguang Deng

Nanyang Technological University, PDCL Lab Sep. 2022 - Sep. 2023
Visiting Ph.D. Student; Host: Prof. Xueyan Tang

Wuhan University of Technology 2015 - 2019
B.Eng.

Research Interests

Learning-augmented online algorithms; intelligent computing systems; agentic services; generative AI systems; cloud-edge intelligence; service computing and resource scheduling.

Selected Honors

- Ningbo Yongjiang Talent (2026)
- Huawei Spark Award (2026)
- REMS 2026 Outstanding Paper Award
- Zhejiang Provincial Outstanding Doctoral Dissertation Nomination (2026)
- SICI Best Paper Award (2025)
- CCF Service Computing Doctoral Dissertation Incentive Program (2025)
- ZJU Outstanding Young Scholar (2024)
- Zhejiang University Outstanding Doctoral Dissertation (2024)
- IEEE ICWS Best Student Paper Award (2019)

Selected Research Projects

- **PI**, National Key R&D Program Project: Adaptive multimodal data-governance technologies for fully connected systems, 2024-2027.
- **Subproject Lead**, National Science and Technology Major Project: Development and application of an operating system for neuromorphic brain-inspired computers, 2026-2029.
- **Task Lead**, NSFC Key Project: Theory and methods for building intelligent service systems across scenarios, 2027-2030.
- **PI**, NSFC Youth Fund: Service scheduling and resource-management optimization through hardware-software collaboration, 2026-2028.
- **Participant**, Zhejiang Provincial Natural Science Foundation Major Project: Resource optimization and computing acceleration for generative AI on heterogeneous hardware, 2025-2027.
- **PI**, Ningbo Yongjiang Talent Programme (Young Talent): Key technologies for collaborative optimization of multi-agent systems in complex open environments, 2026-2031.
- **Participant**, Zhejiang Provincial Science and Technology Program: Digital management and non-invasive diagnosis technologies for heart failure based on EMAT, 2025-2026.
- **Project Lead and Management Committee Member**, Zhejiang University-Baweitong Joint R&D Center for Big Data and Artificial Intelligence: Key technologies for spatial intelligence in industrial applications of embodied AI, 2026-2028.
- **PI**, Health-AI Collaboration: Cranial-nerve health assessment based on artificial-intelligence vision algorithms, 2026-2028.
- **Participant**, Alibaba Collaboration: Large-model load balancing and related inference optimization problems, 2026-2027.
- **Participant**, ByteDance Collaboration: Microservice performance analysis and resource scheduling for heterogeneous clusters, 2025-2026.
- **PI**, Kuaishou Collaboration: Improving the throughput of representative GPU services in Kuaishou inference scenarios, 2024-2025.

Professional Service

Young Editorial Board Member, Computer Engineering & Science; PC member and reviewer for NeurIPS, IJCAI-ECAI, AAAI, PAKDD, and related venues; reviewer for TSC, TMC, FGCS, TKDD, IoTJ, and others.

Publications

Includes first-author, co-first-author, and corresponding-author papers, and papers with Shuiguang Deng as first author and Hailiang Zhao as second author. † Equal contribution; * corresponding author.

Full publication record: Google Scholar

2026

1. Wenzhuo Qian, **Hailiang Zhao***, Ziqi Wang, Zhipeng Gao, Jiayi Chen, Zhiwei Ling, and Shuiguang Deng*. ARMOR: A Robust Self-Supervised Framework for Root Cause Analysis in Microservices under Missing Modality. *ASE*, 2026.
2. Peng Chen, **Hailiang Zhao***, Xueyan Tang, Yixuan Wang, and Shuiguang Deng*. Towards Optimal Robustness in Learning-Augmented Paging. *ICML*, 2026. **Spotlight**.
3. Jiaji Zhang, Ruichao Sun, **Hailiang Zhao***, Jiaju Wu, Peng Chen, Hao Li, Yuying Liu, Kingsum Chow, Gang Xiong, and Shuiguang Deng*. SegQuant: A Semantics-Aware and Generalizable Quantization Framework for Diffusion Models. *CVPR*, 2026.
4. Yinuo Deng†, **Hailiang Zhao†***, Dongjing Wang, Peng Chen, Wenzhuo Qian, Jianwei Yin, Schahram Dustdar, and Shuiguang Deng*. PeerSync: Accelerating Containerized Model Inference at the Network Edge. *IEEE Transactions on Services Computing*, 2026.
5. **Hailiang Zhao**, Xueyan Tang, Peng Chen, Zenong Ye, Tianjun Zhu, Jianwei Yin, and Shuiguang Deng. Learning-Augmented Algorithms: Guarantees, Construction Mechanisms, and System-Level Implications. *arXiv preprint*, 2026.
6. **Hailiang Zhao**, Ziqi Wang, Yifei Zhang, Mingyi Liu, Xinkui Zhao, Kingsum Chow, and Shuiguang Deng. Prediction-Robust Service Deployment with Capacity-Aware Edge Admission. *arXiv preprint*, 2026.
7. Jiaji Zhang, **Hailiang Zhao***, Jiaju Wu, Ruichao Sun, Xinkui Zhao, and Shuiguang Deng*. Shiva-DiT: Residual-Based Differentiable Top-k Selection for Efficient Diffusion Transformers. *arXiv preprint*, 2026.
8. Zhiwei Ling, **Hailiang Zhao***, Chao Zhang, Xiang Ao, Ziqi Wang, Cheng Zhang, Zhen Qin, Xinkui Zhao, Kingsum Chow*, Yuanqing Wu, and MengChu Zhou. Adaptive Dual-Weighting Framework for Federated Learning via Out-of-Distribution Detection. *arXiv preprint*, 2026.
9. Yu Tang†, **Hailiang Zhao†***, Chuansheng Lu, Yifei Zhang, Kingsum Chow*, Shuiguang Deng, and Rui Shi. Morphis: SLO-Aware Resource Scheduling for Microservices with Time-Varying Call Graphs. *arXiv preprint*, 2026.
10. **Hailiang Zhao**, Ziqi Wang, Daojiang Hu, Mingyi Liu, Jiahui Zhai, Kai Di, Xinkui Zhao, Zhongjie Wang, Jianwei Yin, Albert Y. Zomaya, MengChu Zhou, and Shuiguang Deng. Industrial Data-Service-Knowledge Governance: Toward Integrated and Trusted Intelligence. *arXiv preprint*, 2026.

2025

11. Peng Chen, **Hailiang Zhao***, Jiaji Zhang, Xueyan Tang, Yixuan Wang, and Shuiguang Deng*. Robustifying Learning-Augmented Caching Efficiently without Compromising 1-Consistency. *NeurIPS*, 2025.
12. Zhiwei Ling, Yachen Chang, **Hailiang Zhao***, Xinkui Zhao, Kingsum Chow*, and Shuiguang Deng. CADRef: Robust Out-of-Distribution Detection via Class-Aware Decoupled Relative Feature Leveraging. *CVPR*, 2025.
13. **Hailiang Zhao**, Ziqi Wang, Guanjie Cheng*, Wenzhuo Qian, Peng Chen, Jianwei Yin, Schahram Dustdar, and Shuiguang Deng*. Online Workload Scheduling for Social Welfare Maximization in the Computing Continuum. *IEEE Transactions on Services Computing*, 2025.
14. Cheng Zhang, Yinuo Deng, **Hailiang Zhao***, Tianlv Chen, and Shuiguang Deng*. Tail-Learning: Adaptive Learning Method for Mitigating Tail Latency in Autonomous Edge Systems. *ACM Transactions on Autonomous and Adaptive Systems*, 2025.
15. **Hailiang Zhao**, Xueyan Tang*, Peng Chen, Jianwei Yin, and Shuiguang Deng. Data-Locality-Aware Task Assignment and Scheduling for Distributed Job Executions. *IEEE Transactions on Services Computing*, 2025.
16. Shuiguang Deng*, **Hailiang Zhao***, Ziqi Wang, Wenzhuo Qian, Xiang Ao, Guanjie Cheng, Jianwei Yin, Albert Y. Zomaya, and Schahram Dustdar. Agentic Services Computing. *arXiv preprint*, 2025.
17. Jiahong Yu†, Ziqi Wang†, **Hailiang Zhao***, Wei Zhai, Xueqiang Yan, and Shuiguang Deng. TrackTeller: Temporal Multimodal 3D Grounding for Behavior-Dependent Object References. *arXiv preprint*, 2025.
18. Ziqi Wang, **Hailiang Zhao***, Yuhao Yang, Daojiang Hu, Cheng Bao, Mingyi Liu, Kai Di, Schahram Dustdar, Zhongjie Wang, and Shuiguang Deng. OmniFuser: Adaptive Multimodal Fusion for Service-Oriented Predictive Maintenance. *arXiv preprint*, 2025.
19. Peng Chen, Jiaji Zhang, **Hailiang Zhao***, Yirong Zhang, Shenyao Chen, Jiahong Yu, Xueyan Tang, Yixuan Wang, Hao Li, Jianping Zou, Gang Xiong, Kingsum Chow, Shuibing He, and Shuiguang Deng*. Toward Robust and Efficient ML-Based GPU Caching for Modern Inference. *arXiv preprint*, 2025.
20. Ziqi Wang, **Hailiang Zhao***, Cheng Bao, Wenzhuo Qian, Yuhao Yang, Xueqiang Sun, and Shuiguang Deng*. XFMNet: Decoding Cross-Site and Nonstationary Water Patterns via Stepwise Multimodal Fusion for Long-Term Water Quality Forecasting. *arXiv preprint*, 2025.
21. Wenzhuo Qian, **Hailiang Zhao***, Jiayi Chen, Ziqi Wang, Tianlv Chen, Zhiwei Ling, Xinkui Zhao, Kingsum Chow, Albert Y. Zomaya, and Shuiguang Deng*. Reliable Microservice Tail Latency Prediction via Decoupled Dual-Stream Learning and Gradient Modulation. *arXiv preprint*, 2025.

2024

22. **Hailiang Zhao**, Xueyan Tang[*], Peng Chen, and Shuiguang Deng. Learning-Augmented Algorithms for the Bahncard Problem. *NeurIPS*, 2024.
23. Shuiguang Deng[*], **Hailiang Zhao**[*], Binbin Huang, Cheng Zhang, Feiyi Chen, Yinuo Deng, Jianwei Yin, Schahram Dustdar, and Albert Y. Zomaya. Cloud-Native Computing: A Survey From the Perspective of Services. *Proceedings of the IEEE*, 2024. **Cover Paper**.
24. **Hailiang Zhao**, Shuiguang Deng[*], Zhengzhe Xiang, Xueqiang Yan, Jianwei Yin, Schahram Dustdar, and Albert Y. Zomaya. Scheduling Multi-Server Jobs With Sublinear Regrets via Online Learning. *IEEE Transactions on Services Computing*, 2024.

2023

25. **Hailiang Zhao**, Shuiguang Deng[*], Feiyi Chen, Jianwei Yin, Schahram Dustdar, and Albert Y. Zomaya. Learning to Schedule Multi-Server Jobs With Fluctuated Processing Speeds. *IEEE Transactions on Parallel and Distributed Systems*, 2023.

2022

26. Shuiguang Deng, **Hailiang Zhao**, Zhengzhe Xiang, Cheng Zhang, Rong Jiang, Ying Li[*], Jianwei Yin, Schahram Dustdar, and Albert Y. Zomaya. Dependent Function Embedding for Distributed Serverless Edge Computing. *IEEE Transactions on Parallel and Distributed Systems*, 2022.
27. **Hailiang Zhao**, Shuiguang Deng[*], Zijie Liu, Zhengzhe Xiang, Jianwei Yin, Schahram Dustdar, and Albert Y. Zomaya. DPoS: Decentralized, Privacy-Preserving, and Low-Complexity Online Slicing for Multi-Tenant Networks. *IEEE Transactions on Mobile Computing*, 2022.
28. **Hailiang Zhao**, Shuiguang Deng[*], Zijie Liu, Jianwei Yin, and Schahram Dustdar. Distributed Redundant Placement for Microservice-based Applications at the Edge. *IEEE Transactions on Services Computing*, 2022. **ESI Highly Cited Paper**.

2020

29. Shuiguang Deng, **Hailiang Zhao**, Weijia Fang[*], Jianwei Yin, Schahram Dustdar, and Albert Y. Zomaya. Edge Intelligence: The Confluence of Edge Computing and Artificial Intelligence. *IEEE Internet of Things Journal*, 2020. **ESI Hot & Highly Cited Paper**.

2019

30. **Hailiang Zhao**, Shuiguang Deng[*], Cheng Zhang, Wei Du, Qiang He, and Jianwei Yin. A Mobility-Aware Cross-Edge Computation Offloading Framework for Partitionable Applications. *IEEE ICWS*, 2019. **Best Student Paper Award**.

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